Individual and Market Demand

Chapters 5

1 How Income Changes Affect an Individual's Consumption Choices

> **Income Effect:** The change in a consumer's consumption choices that results from a change in the purchasing power of the consumer's income.

Figure 5.1 shows the effect of an increase in income on consumption for Eve, a consumer who allocates her income between vacations and tickets to basketball games. Her initial budget constraint BC1 is tangent to the indifference curve U1 at the optimal consumption bundle A. An increase in Eve's income is represented by the outward parallel shift of BC1 to BC2. Because the prices of the goods are unchanged, Eve can now afford to buy more vacations and basketball tickets, and her new utility-maximizing bundle is B, where indifference curve U2 is tangent to BC2.

Suppose that the individual's income has increased.

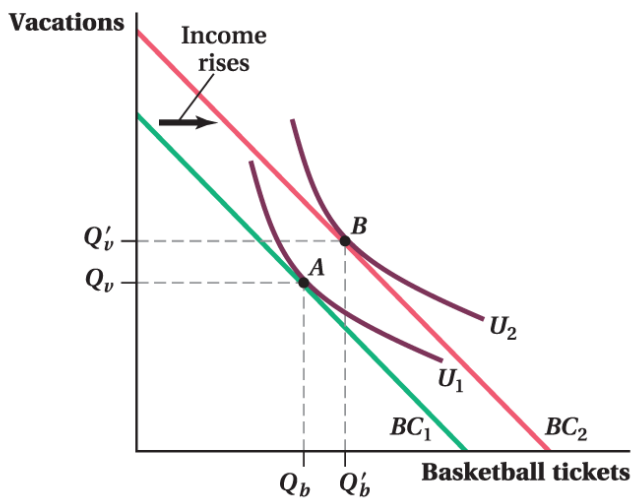

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Figure 1

1.1 Normal and Inferior Goods

> **Normal Good:** A good for which consumption rises when income rises.

> **Inferior Good:** A good for which consumption decreases when income rises.

Figure 5.2 presents an example in which one of the goods, macaroni and cheese, is inferior.

Suppose income has increased:

- We can see that Instant Ramen is an inferior good because the quantity purchased fell from R_1 to R_2.
- We can see that steak is a normal good because the quantity purchased increased from S_1 to S_2.

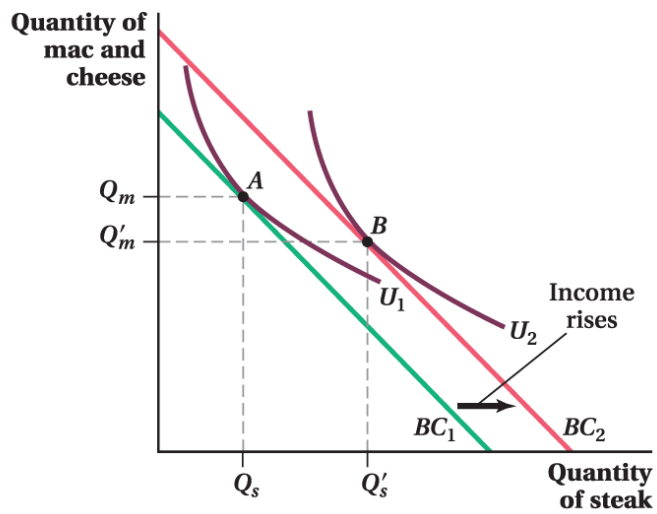

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Figure 2

What kinds of goods tend to be inferior? Usually, they are lower-quality or otherwise less-desirable goods.

1.2 Income Elasticities and Types of Goods

We've discussed how the income effect can be positive (as with normal goods) or negative (as with inferior goods). We can make further distinctions between types of goods by looking not just at the sign of the income effect, but at the income elasticity as well, which we discussed in Chapter 2.

> **Income Elasticity:** The percentage change in the quantity consumed of a good in response to a given percentage change in income.

$$E_I^D = \frac{\%\Delta Q}{\%\Delta I} = \frac{\Delta Q/Q}{\Delta I/I} = \frac{\Delta Q}{\Delta I}\frac{I}{Q}$$

The first ratio in the income elasticity definition is the income effect shown in the equations above: $\Delta Q/\Delta I$, the change in quantity consumed in response to a change in income. Positive for normal goods, negative for inferior goods.

> **Necessity Good:** A normal good for which income elasticity is between zero and 1.

> **Luxury Good:** A good with an income elasticity greater than 1.

POLL EVERYWHERE: A consumer's bundle includes good X and good Y. If good X is a normal good, goods X and Y are substitutes, and the price of good X rises, the consumer will buy _____ of good X and _____ of good Y.

1.3 The Income Expansion Path

Imagine repeating the analysis in the previous section for every possible income level, starting with 0. That is, for a given set of prices and a particular set of preferences, find the utility-maximizing bundle for every possible budget constraint, where each constraint corresponds to a different income level.

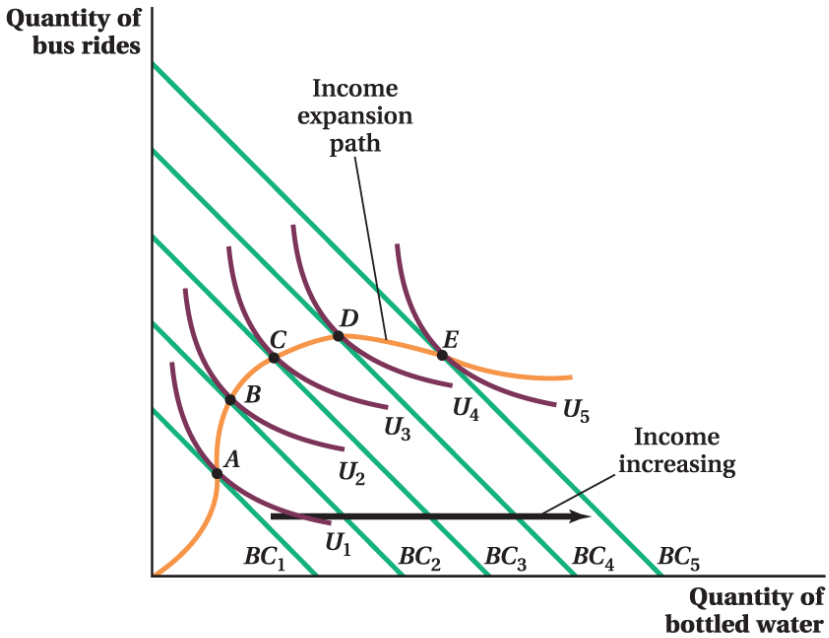

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Figure 3

> **Income Expansion Path:** A curve that connects a consumer's optimal bundles at each income level.

When both goods are normal goods, the income expansion path will be positively sloped because consumption of both goods rises when income does. If the slope of the income expansion path is negative, then the quantity consumed of one of the goods falls with income while the other rises. The one whose quantity falls is an inferior good

1.4 The Engel Curve

The income expansion shows how consumer behavior changes in response to changes in income, but it has two important weaknesses:

1. with just two axes, we can only look at two goods at a time.
2. although we can easily see the consumption quantities of each good, we can't discern the income level to which a particular point on the curve corresponds.

A better way to see how the quantity consumed of one good varies with income (as opposed to how the relative quantities of the two goods vary) is to take the information conveyed by the income expansion path and plot it on a graph with income on the vertical axis and the quantity of the good in question on the horizontal axis.

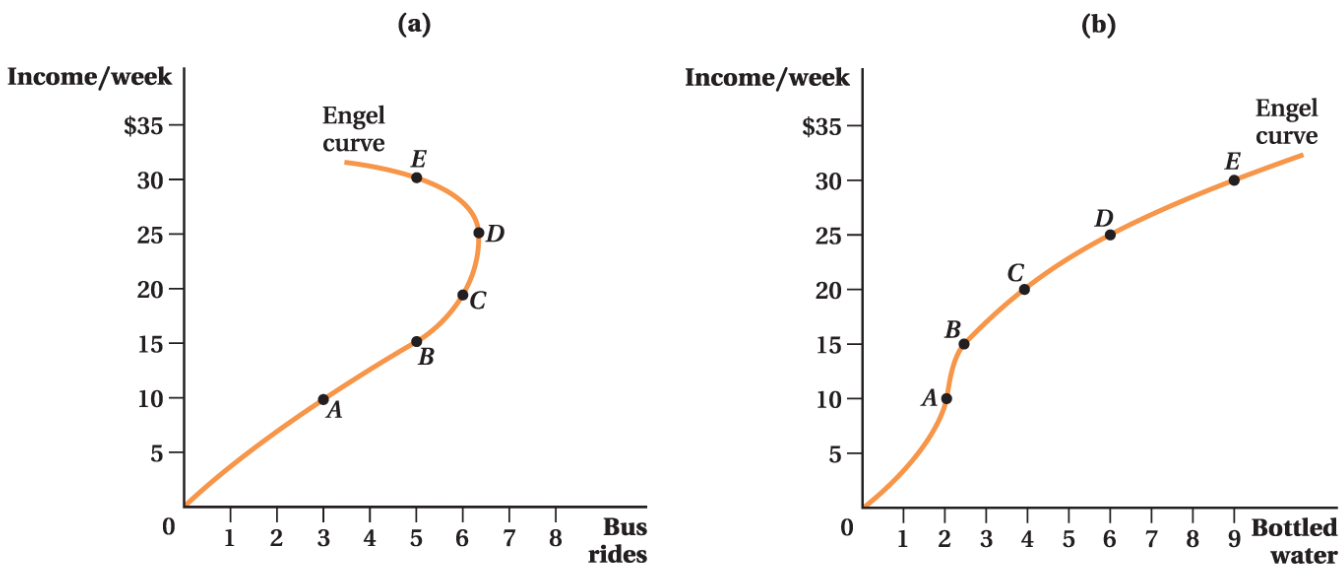

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Figure 4

> **Engel Curve:** A curve that shows the relationship between the quantity of a good consumed and a consumer's income.

2 How Price Changes Affect Consumption Choices

2.1 Deriving a Demand Curve

Caroline is deciding how to spend her income on two goods, 2-liter bottles of Mountain Dew and 1-liter bottles of grape juice, and we want to know her demand curve for grape juice. Caroline's income is $20, and the price of Mountain Dew is $2 per 2-liter bottle.

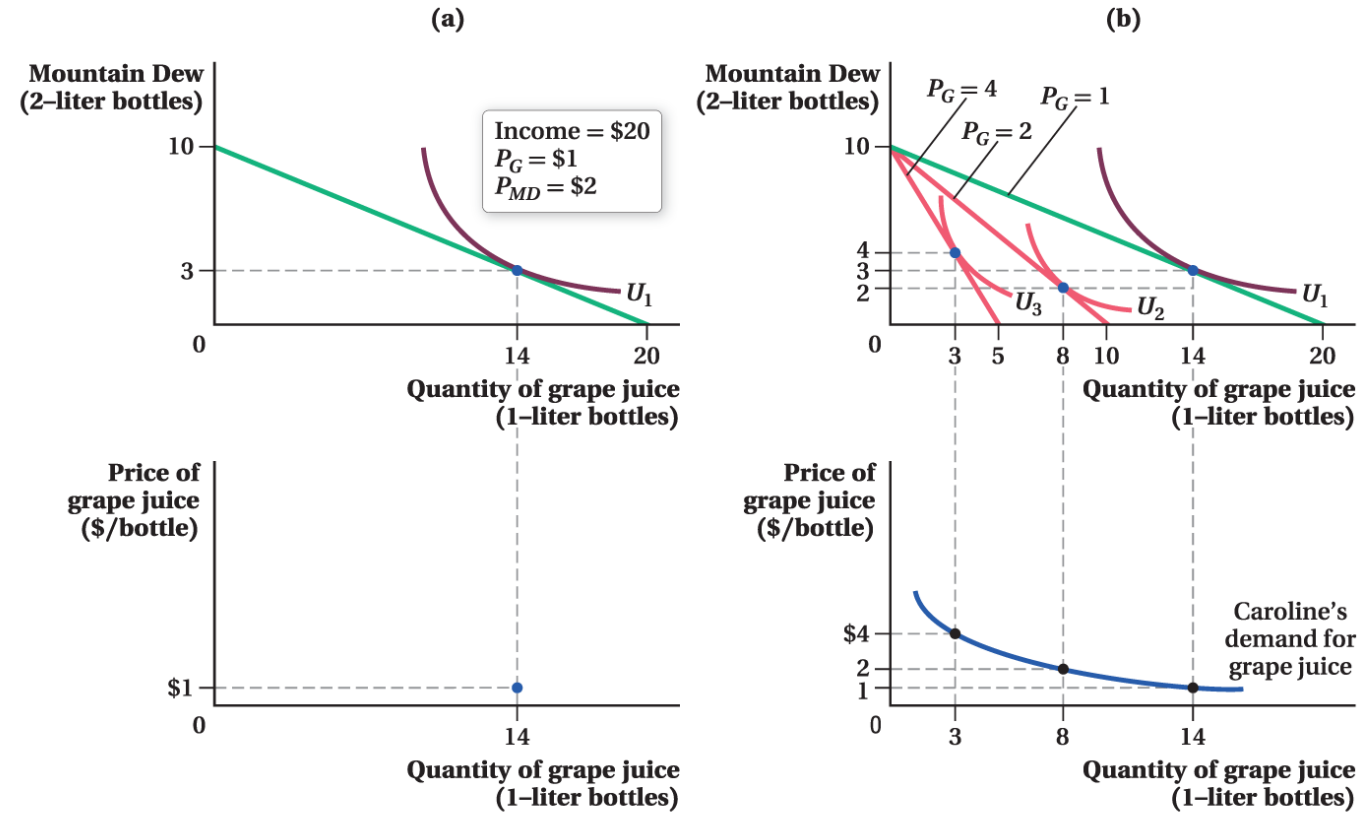

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Figure 5

2.2 Shifts in the Demand Curve

We know that if a consumer's preferences or income change, or the prices of other goods change, then the demand curve shifts. We can see how by tracing out the demand curve under these new conditions.

Let's look at an example where preferences change. Suppose that Caroline meets a scientist at a party who argues that the purported health benefits of grape juice are overstated and that it stains your teeth red. This changes Caroline's preferences toward grape juice. She finds it less desirable.

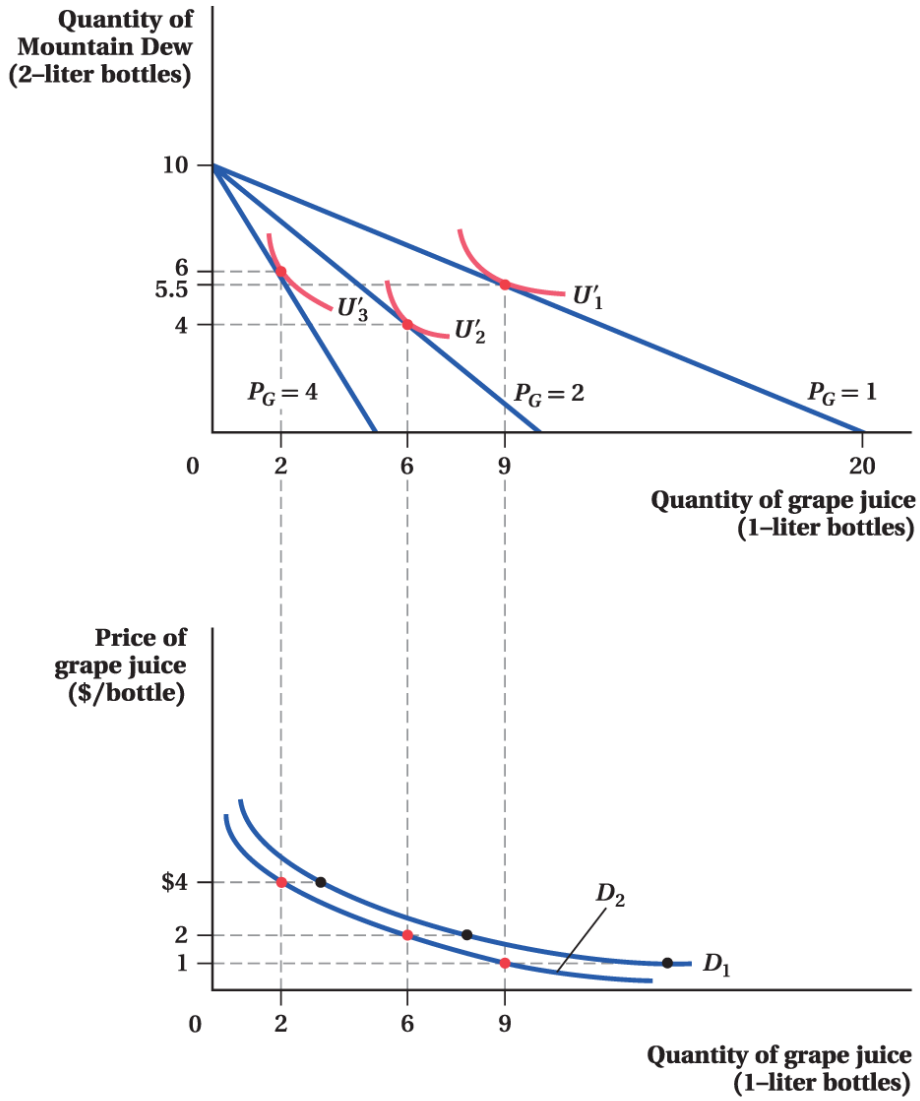

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Figure 6

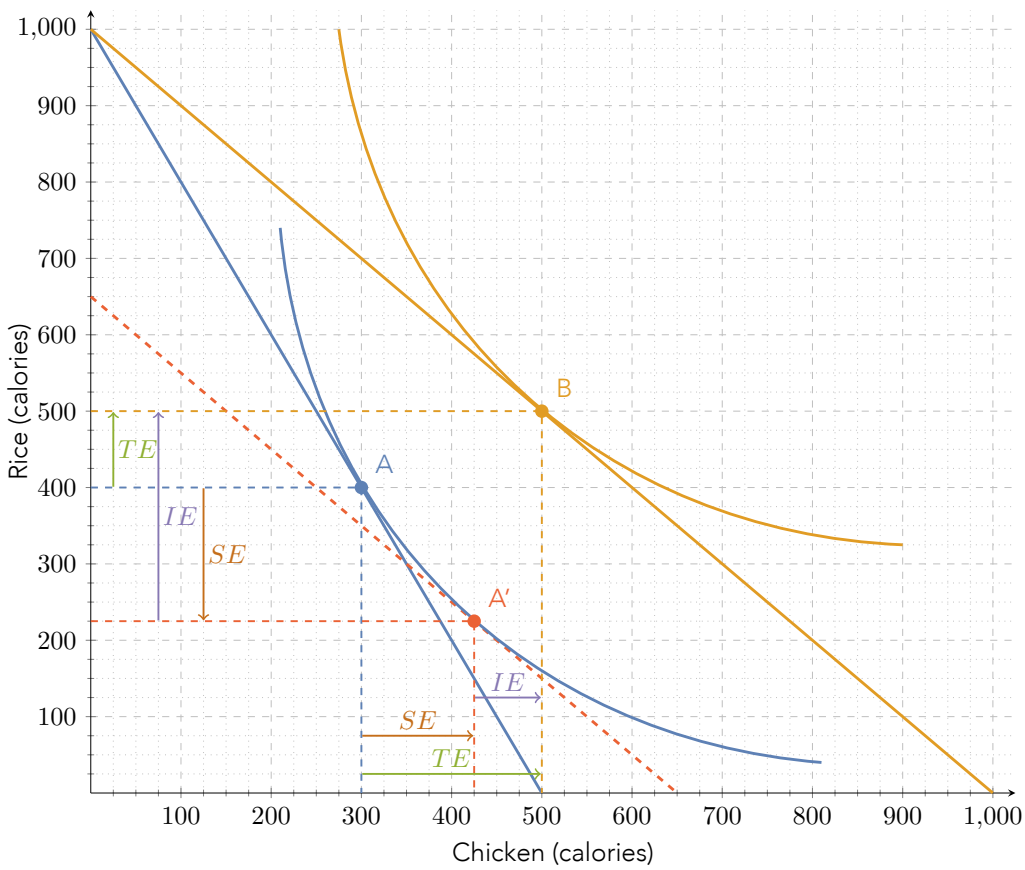
3 Consumer Response to Price Changes: Substitution and Income Effects

> **Substitution Effect:** The change in a consumer's consumption choices that results from a change in the relative prices of two goods.

> **Income Effect:** The change in a consumer's consumption choices that results from a change in the purchasing power of the consumer's income.

> **Total Effect:** the sum of the income and substitution effects.

- A household going to eat a dinner of chicken and rice and has $20 to spend. Currently, each calorie of rice costs $0.02 and each calorie of chicken costs $0.04.
  - The optimal consumption bundle is point A and the household will consume 400 calories of rice and 300 calories of chicken .
- Suppose that the price of chicken falls to $0.02 per calorie.
  - The optimal consumption bundle is now point B and the household will consume 500 calories of rice and 500 calories of chicken.
- Consumption bundle A' would be the optimal choice for the household if the faced the new prices after the change but maintained the same level of utility that they had before the change.
  - In other words, it is what the households would have chosen after the price change if they did not feel like their real income changed.



- Total Effect: the change in the consumption of each good as a result of the price decrease ($A \to B$).
  - Total Effect for Rice (TE_R): $500 - 400 = 100$.

  - Total Effect for Chicken (TE_C): $500 - 300 = 200$.

- Income Effect: the increased real income that results from the price decrease.
  - The original bundle (A) now only costs: $0.02 \times 400 + 0.02 \times 300 = 8 + 6 = 14$
  - The household can now buy more of each good because the price of chicken has gone down.
    - The income effect is positive for both goods (the fall in the price causes you to buy more of both goods).
  - The income effect is shown by the movement from $A' \to B$.
    - Income effect for rice: $500 - 225 = 275$
    - Income effect for chicken: $500 - 425 = 75$
- Substitution Effect: The price decrease for chicken makes rice relatively more expensive and chicken relatively cheaper.
  - The substitution effect for rice is negative causing you to buy less rice.
  - The substitution effect for chicken is positive causing you to buy more chicken.
  - The substitution effect is shown by the movement from $A \to A'$.
    - Substitution effect for rice: $225 - 400 = -175$
    - Substitution effect for chicken: $425 - 300 = 125$.

A table of income and substitution effects:

Price of Good X Goes Up			Price of Good X Goes Down		
	Good X	Good Y		Good X	Good Y
Substitution Effect			Substitution Effect		
Income Effect			Income Effect		
Price of Good Y Goes Up			**Price of Good Y Goes Down**		
	Good X	Good Y		Good X	Good Y
Substitution Effect			Substitution Effect		
Income Effect			Income Effect		
Income Goes Up			**Income Goes Down**		
	Good X	Good Y		Good X	Good Y
Substitution Effect			Substitution Effect		
Income Effect			Income Effect		

An Example with Numbers: Suppose that a consumer is consuming at their optimal point (A) and is consuming $X = 10$ and $Y = 5$. Then the price of good Y falls and the consumer's optimal consumption bundle (B) changes to $X = 8$ and $Y = 9$. For each of the goods, can we tell which is bigger (in absolute value) the income effect or the substitution effect?

- Total Effect:
  - Total effect for good X: $8 - 10 = -2$.
  - Total Effect for good Y: $9 - 5 = 4$.
- Substitution Effect:
  - Good X is now relatively more expensive, causing people to buy less of good X, which means the substitution effect for good X is negative.
  - Good Y is now relatively cheaper, causing people to buy more of good Y, which means the substitution effect for good Y is positive.
- Income Effect:
  - The price of good Y fell, so real income has increased, causing people to buy more of good X, which means the income effect for good X is positive.
  - The price of good Y fall, so real income has increased, causing people to buy more of good Y, which means the income effect for good Y is positive.
- Comparing the two effects
  - Good X:
    - The substitution effect causes people to buy less of good X.
    - The income effect causes people to buy more of good X.
    - Since the total effect is negative (overall people are buying less of good X) that means the substitution effect is stronger than the income effect.
  - Good Y
    - The substitution effect causes people to buy more of good Y.
    - The income effect causes people to buy more of good Y.
    - Since the total effect is positive (overall people are buying more of good Y) we cannot tell which effect is stronger without more information because both the substitution and income effects are causing people to buy more of good Y.

3.1 What Determines the Size of the Substitution and Income Effects

3.1.1 The Size of the Substitution Effect

The size of the substitution effect depends on the degree of curvature of the indifference curves. This makes sense curvier indifference curves mean the goods are complements, flatter means substitutes.

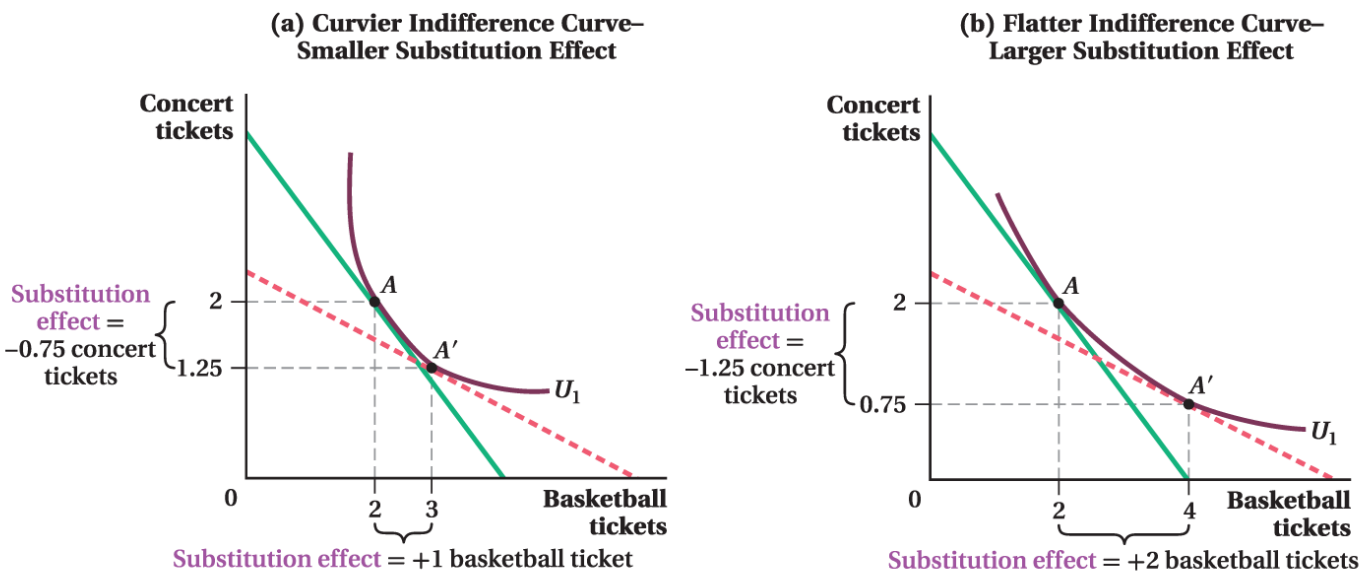


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Figure 7

3.1.2 The Size of the Income Effect

The size of the income effect is related to the quantity of each good the consumer purchases before the price change. The more the consumer was spending on the good before the price change, the greater the fraction of the consumer's budget affected by the price change.

3.2 An Example of the Income and Substitution Effects with an Inferior Good

Suppose there is a fall in the price of Ramen Noodles

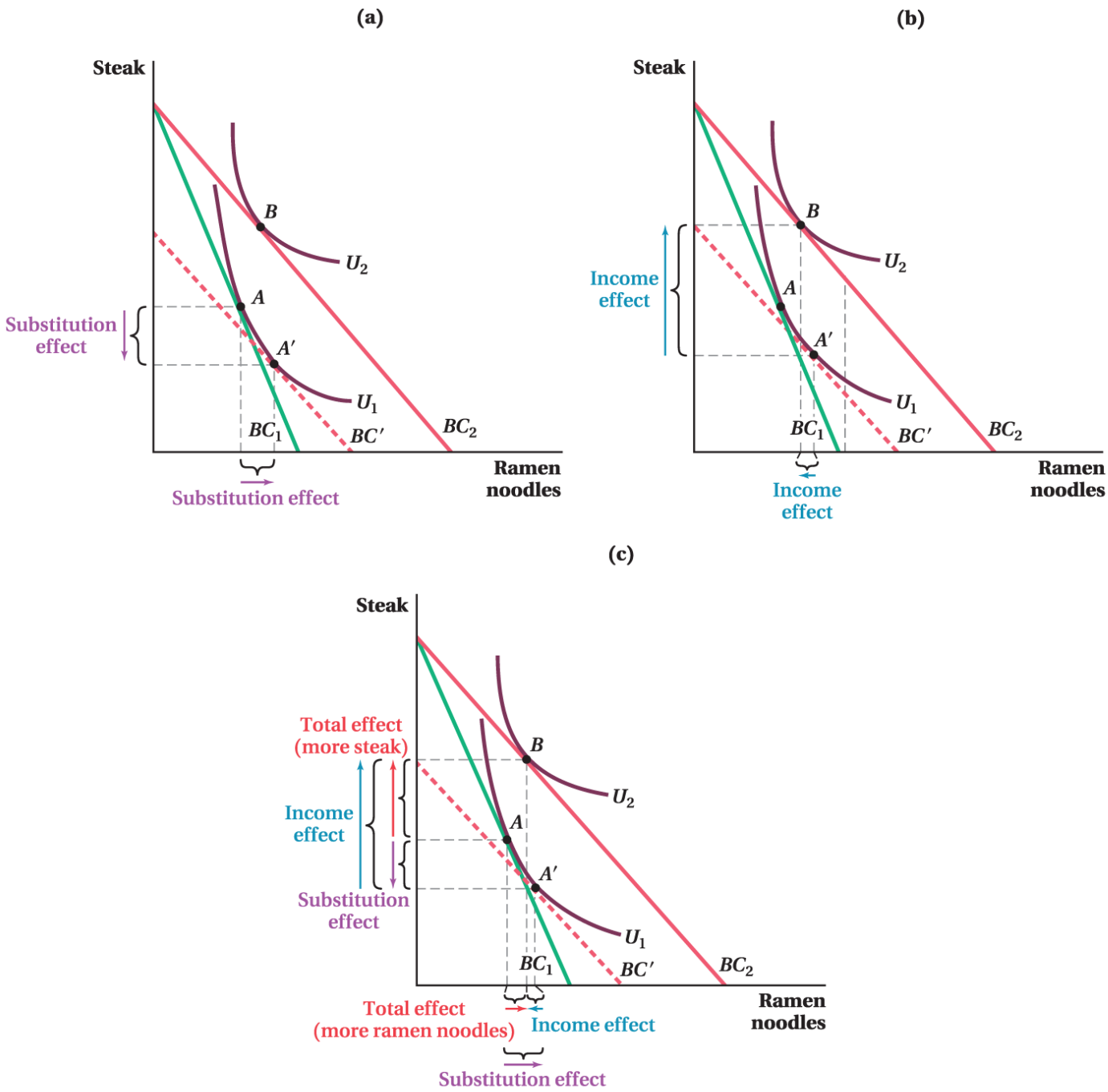

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Figure 8

3.3 Giffen Goods

> **Giffen Goods:** A good for which a fall in price leads the consumer to want less of the good.

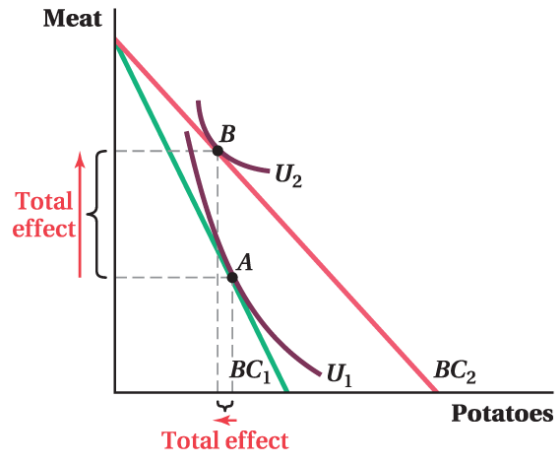

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Figure 9

4 The Impact of Changes in Another Good's Price; Substitutes and Complements

4.1 A Change in the Price of a Substitute Good

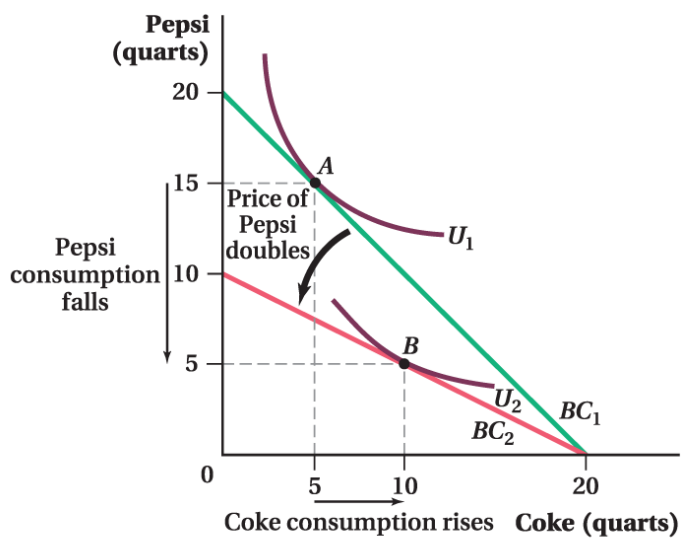

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Figure 10

4.2 A Change in the Price of a Complement Good

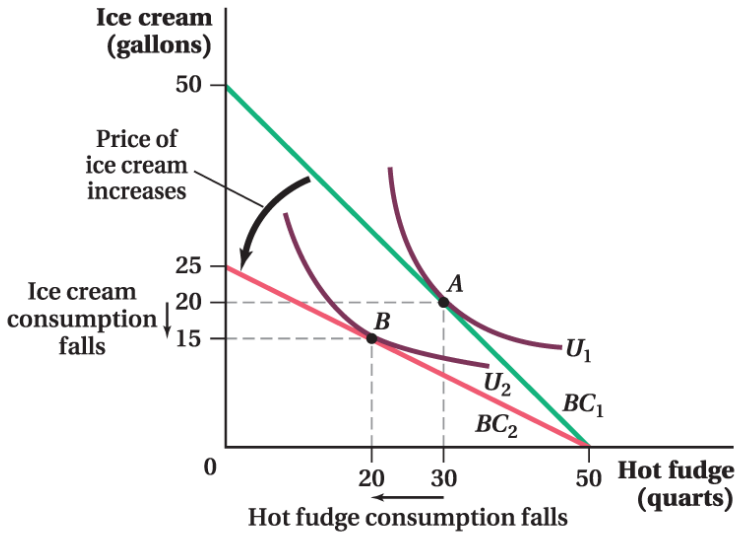

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Figure 11

4.3 Impact on the Demand Curve

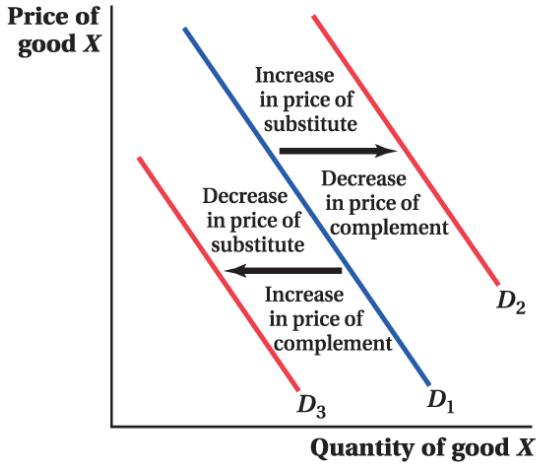

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Figure 12

5 Combining Individual Demand Curves to Obtain the Market Demand Curve

The market demand for a good is the sum of all the individual demand curves. That is, the market quantity demanded for a good at a particular price is the sum of every individual consumer's quantity demanded at that price.

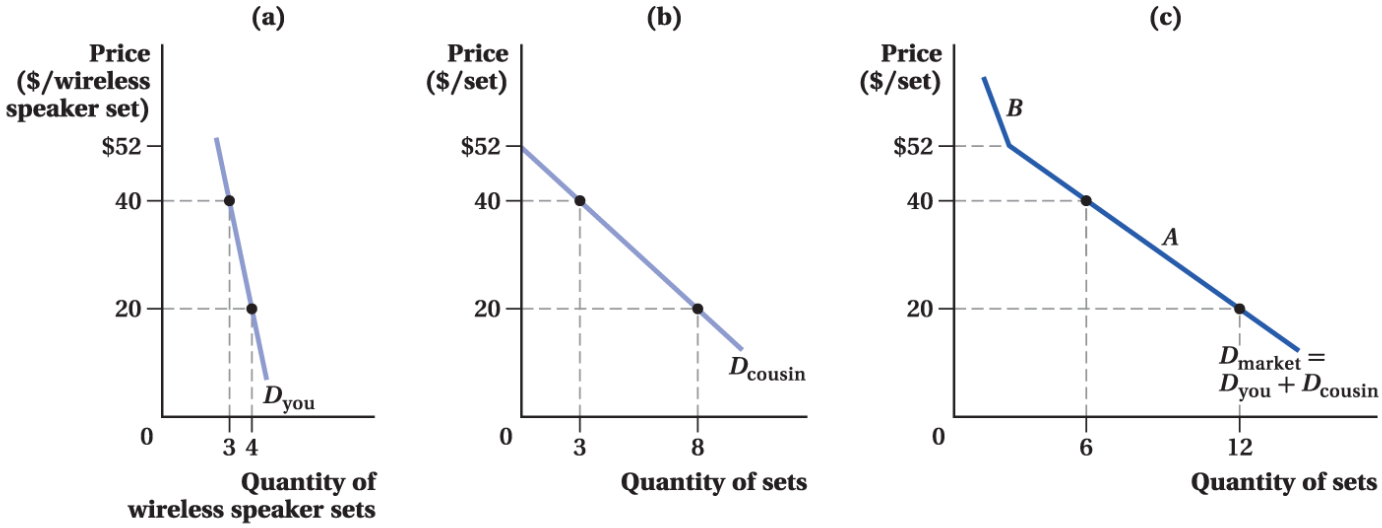

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Figure 13

5.1 Using Algebra to Move from Individual to Market Demand

$$Q_A = 5 - 0.05P$$
$$Q_B = 13 - 0.25P$$
$$Q_M = Q_A + Q_B = 5 - 0.05P + 13 - 0.25P = 18 - 0.3P$$

A and B's choke points are different (A's is $100 and B's is $52), this means for prices above $52 the market demand curve is only A's. There is a kink at $52.

POLL EVERYWHERE: To calculate the market demand curve from individual demand curves: